Huawei Access Node JAY-ACC-35 Hardware/Environmental Fault Resolution SOP

Node: JAY-ACC-35 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_8133

Description

The failure of Node ID JAY-ACC-35, located at the Jayanagar Metro Site, is attributed to hardware or environmental faults within the Huawei Access Node. Specifically, the occurrence of ASIC memory parity errors and line card crashes, coupled with the loss of power supply redundancy, indicate a critical issue requiring prompt attention. This SOP outlines the steps to diagnose and resolve the issue, focusing on the role of the Access Node and the Huawei CLI dialect. The underlying protocol and hardware mechanics contributing to this failure domain involve the interaction between the Node's control and data planes. The control plane, responsible for routing and signaling, is susceptible to faults caused by hardware malfunctions, such as ASIC memory parity errors. Meanwhile, the data plane, handling packet forwarding, is impacted by line card crashes and power supply redundancy loss. By executing the commands outlined in this SOP, network engineers can identify the root cause of the failure and implement corrective actions to restore the Node's functionality.

Pre-requisite Checks

- Verify that the Jayanagar Metro Site is accessible and that dispatch has been notified.
- Check the redundant link status to ensure that the Node's connectivity is maintained.
- Gather information on the failure time and any relevant logs.

Potential Root Causes

Cause	Probability
ASIC memory parity errors	High
Line card crashes	Medium
Power supply redundancy loss	Low

Diagnostic Commands & Expected Outputs

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Command: display cpu-usage, Expected Output: CPU usage: 90%, Description: This command displays the CPU usage of the Node, helping to identify if the ASIC memory parity errors are causing high CPU utilization.

Command: display board-info, Expected Output: Board ID: 1, Board Type: Line Card, Description: This command displays the board information, including the board ID and type. It helps to identify if the line card crashes are occurring on the affected board.

Command: display power-supply, Expected Output: Power Supply Status: Offline, Description: This command displays the power supply status,
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helping to identify if the power supply redundancy loss is the root cause of the failure.
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Resolution Steps

Step 1: Step: Replace the faulty ASIC, Command: undo cpu-asic replace, Description: This command replaces the faulty ASIC with a new one, resolving the ASIC memory parity errors.

Step 2: Step: Replace the crashed line card, Command: undo board replace, Description: This command replaces the crashed line card with a new one, resolving the line card crashes.

Step 3: Step: Restore power supply redundancy, Command: undo power-supply restore, Description: This command restores the power supply redundancy, resolving the power supply redundancy loss.

Post-Check Verification

```
Command: display cpu-usage, Expected Output: CPU usage: Normal, Description: This command verifies that the CPU usage is normal after the resolution steps.

Command: display board-info, Expected Output: Board ID: 1, Board Type: Line Card, Status: Online, Description: This command verifies that the line card is online after the replacement.

Command: display power-supply, Expected Output: Power Supply Status: Online, Description: This command verifies that the power supply redundancy is restored.
```

Roll-back Procedures

- Step: Restore the original ASIC, Command: cpu-asic replace undo, Description: This command restores the original ASIC after the replacement failed.

- Step: Restore the original line card, Command: board replace undo, Description: This command restores the original line card after the replacement failed.

- Step: Restore the original power supply status, Command: power-supply restore undo, Description: This command restores the original power supply status after the restoration failed.